Unit - 2

Question:
Explain the introduction and role of Vision Systems in Manufacturing.

Answer:

Vision Systems are integrated components that utilize information from digital images.
They guide manufacturing and production operations such as:

- Go/no-go testing
- Quality control processes

They support automated assembly verification and inspection by: Guiding material handling equipment to correctly position products or materials
<u>Applications:</u>

- Used across various industries
- Automate mundane and repetitive tasks that would exhaust human inspectors

<u>Benefits of machine vision systems:</u>

- Enable 100% inspection of products or parts
- Improve yield
- Reduce defect rates
- Increase product quality
- Lower manufacturing costs
- Ensure consistent process results

<u>Common manufacturing uses:</u>

- Product and quality inspection
- Structure surveillance
- Damage or fault tracking

<u>Advantage over human inspection:</u>

- Cameras can detect tiny defects
- Machines remain consistently attentive and do not tire

Question:
Explain the role of human vision in computer interaction in manufacturing.

Answer:

Vision-based human-computer interaction enables a broader and more expressive range of input capabilities.
It uses computer vision techniques to process data from one or more cameras.

- Human–Computer Interaction (HCI):
    - A research area focused on the design and use of computer technology.
    - Concentrates on the interface between people (users) and computers.
    - Researchers study how humans interact with computers and develop technologies that allow new forms of interaction.

- Human–Machine Interaction (HMI):
    - Refers to communication and interaction between a human and a machine through a user interface.

- Natural user interfaces, such as gesture control, are gaining popularity:
    - They allow humans to interact with machines using intuitive and natural behaviours.

Question:
List and explain the types of computer vision used in manufacturing.

Answer:

The types of computer vision in manufacturing include:

<u>Image segmentation:</u>

- Converts an image into a collection of pixel regions represented by a mask or labelled image.
- Helps in processing only important segments instead of the entire image.

<u>Object detection:</u>

- Involves gathering and organizing data.
- The system interprets pixel-based or vector-based images, identifies important data, and classifies it into distinct objects.

<u>Edge detection:</u>

- An image processing technique used to find object boundaries within images.
- Works by detecting brightness discontinuities.
- Used for segmentation and data extraction in computer vision and machine vision.

<u>Pattern detection / Pattern recognition:</u>

- A data analysis method using machine learning to automatically identify patterns and regularities.
- Can process various data types like text, images, sounds, etc.
- Recognizes familiar patterns quickly and accurately.

<u>Image classification:</u>

- Applied in automated inspection systems through industrial image processing.
- Enhances image quality and desired features for pattern recognition.

- Two methods used:
    - Analogue image processing (for hard copies like printouts and photographs)
    - Digital image processing
- Image analysts use fundamental interpretation techniques during analysis.

<u>Feature matching:</u>

- A manufacturing feature is a group of related geometric elements.
- These elements relate to specific manufacturing methods or are used to determine suitable methods for creating the geometry.

Question:
Differentiate between Artificial Intelligence and Computer Vision in the context of manufacturing.

Answer:

- Computer vision is used to process images based on a set of general rules.
- Artificial Intelligence (AI) is a broader field where machines learn to perform complex tasks independently.
- Example: In object recognition, computer vision applies rule-based processing, whereas AI learns from data.
- Computer vision is a subfield of AI.
- It focuses specifically on enabling machines to interpret content from images or videos.
- AI, in contrast, also includes:
    - Language processing algorithms
    - Prediction algorithms
    - Algorithms that understand dynamic patterns over time
- Computer vision aims to replicate human-like recognition and discrimination of objects, similar to what humans do when walking or watching a movie.
    - Option A: Using hard-coded descriptions of every object (impractical)
    - Option B: Teaching computers to recognize and interact using AI technologies like deep learning
- Computer vision systems, being AI-enabled, can:
    - Recognize images with high accuracy
    - Analyse thousands of products or processes in minutes
    - Perform better than the human brain in terms of speed and consistency

Question:
List and explain the industrial applications of computer vision systems in manufacturing.

Answer:

1. **Automated assembly:**
    - Controls cycle time
    - Enables 3D vision monitoring
    - Optimizes supply chains
    - Supports digital lean manufacturing

2. **Quality inspection of products:**
    - Used for identifying defects and ensuring product quality

3. **Safety:**
    - Tracks objects in videos and detects them in images
    - Enhances industrial security systems

4. **Factory synchronization and dynamic scheduling:**
    - Uses systems like ERP and MES for dynamic resource allocation and scheduling
    - Employs visual data for real-time tracking of materials and products
    - Results in improved agility, reduced inventory, and better equipment effectiveness

5. **Predictive maintenance:**
    - Prevents expensive downtimes by monitoring machinery for early signs of degradation
    - Reinforced with IoT and deep learning for higher accuracy
    - Alerts engineers before failures occur

6. **Reading barcodes and QR codes:**
    - Helps recognize and read barcodes and text in manufacturing environments
    - Used in smart factories for automation with AI and computer vision systems

7. **Training:**
    - Used within machine learning systems to create virtual environments
    - Helps in training and skill development

8. **Track-and-trace:**
    - Tracks products from production to delivery
    - Especially important for sectors like food and medicine manufacturing

Question:
Define welding and classify its types with examples.

Answer:

Welding is a process of joining two pieces of metal by applying heat, with or without pressure and filler material.

It results in a permanent joint.

Common applications include joining: Pressure vessels, Tanks, Bridges, Railways, Machine frames and brackets, Automobile bodies, Aircraft and ship structures.

Welding is broadly classified into:

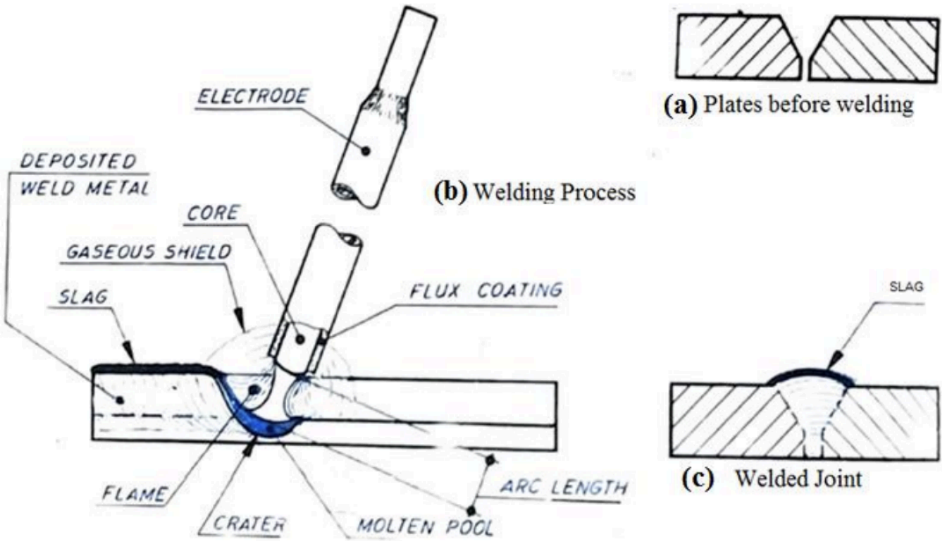
1. Plastic welding or Pressure welding:
    - Metal pieces are heated to a plastic state.
    - Joined by applying pressure without using filler material.
    - Examples:
        - Forge welding
        - Resistance welding
2. Fusion welding or Non-pressure welding:
    - Metal pieces are heated to a molten state and allowed to solidify.
    - Filler material is used in the process.
    - Examples:
        - Arc welding
        - Gas welding

Question:
Explain the arc welding process and list the safety devices used in arc welding.

Answer:

Arc Welding Process:

- When two conductors in an electric circuit are briefly touched and then slightly separated, an electric arc is formed.
- The arc generates intense heat, reaching temperatures of 5000–6000°C along its length.
- The workpiece is connected to one terminal of the power source, and the electrode (held by the operator) to the other.
- The arc melts the metal surface of the workpiece, forming a molten pool directly beneath the arc.
- Simultaneously, the electrode tip also melts, and the molten electrode material is transferred into the molten pool by the electric arc.
- Arc action agitates the molten metal pool, mixing the base metal and filler metal.
- As the molten metal cools, it solidifies to form a strong joint.
- The electrode's flux coating produces an inert gas shield that prevents oxidation of the molten metal by isolating it from atmospheric gases.



Safety Devices Used in Arc Welding:

- Welding shield – protects eyes from infrared and ultraviolet radiation
- Gloves – protect hands from sparks and electrical contact
- Chipping hammer and wire brush – remove slag after welding
- Apron – protects clothing from spatter and sparks
- Earthing clamp – prevents electrical hazards during welding

Question:
Explain the working of an arc welding machine and the types of current used.

Answer:

The welding machine generates low voltage (10–50 V) and high current (50–300 A).

The current can be either AC (alternating current) or DC (direct current), depending on:

- Type of electrode
- Type of metal being welded

AC Arc Welding:

- Uses a step-down transformer to reduce voltage from 220/440 V to 80–100 V
- Current ranges from 100–400 A
- Polarity changes every cycle
- Current and voltage drop to zero twice per cycle
- Requires higher voltage to maintain a stable arc

DC Arc Welding:

- Workpiece is connected to the positive terminal of the DC generator
- Electrode is connected to the negative terminal — this is called **straight polarity**
- Straight polarity is used when high heat is needed
- When less heat is required, polarity is reversed

- DC welding allows polarity reversal, making it suitable for welding metals that require higher heat to melt

Question:
Describe arc welding electrodes and explain the purpose of their flux coating and its ingredients.

Answer:

<u>Arc Welding Electrodes:</u>

- Electrodes used are of consumable type and coated with flux.
- Flux coating materials include:
    - Chalk (lime)
    - Ferro manganese
    - Cellulose
    - Starch
    - Kaolin (China clay)
    - Iron powder
- After welding, the flux forms a slag that must be removed using a chipping hammer and wire brush.

<u>Purposes of Electrode Coating:</u>

- Protects molten metal from oxidation
- Prevents rapid cooling of molten metal
- Helps establish and maintain the arc
- Adds alloying elements to the weld

<u>Purpose of Each Ingredient in Flux Coating:</u>

- China clay, mica:
    - Produce light-weight slag
    - Form a protective layer on molten metal
    - Shield metal from atmospheric contamination
- Cellulose, wood, starch, calcium carbonate:
    - Create a protective gas shield around the electrode, arc, and weld pool
- Ferro manganese (deoxidizer):
    - Refines the molten metal
- Ferro alloys (manganese, molybdenum, etc.):
    - Add desired strength and properties to the weld
- Iron powder:
    - Enhances arc behaviour and bead appearance
    - Increases metal deposition rate and arc travel speed

Question:
Explain the gas welding process and describe the oxy-acetylene welding setup.

Answer:

Gas Welding:

- Gas welding is a fusion welding method where a strong gas flame is used to melt the workpiece.
- Similar to arc welding, a filler material is used to fill the joint.
- Common gas combinations for heating:
    - Oxygen and Acetylene
    - Oxygen and Hydrogen

Oxy-Acetylene Welding:

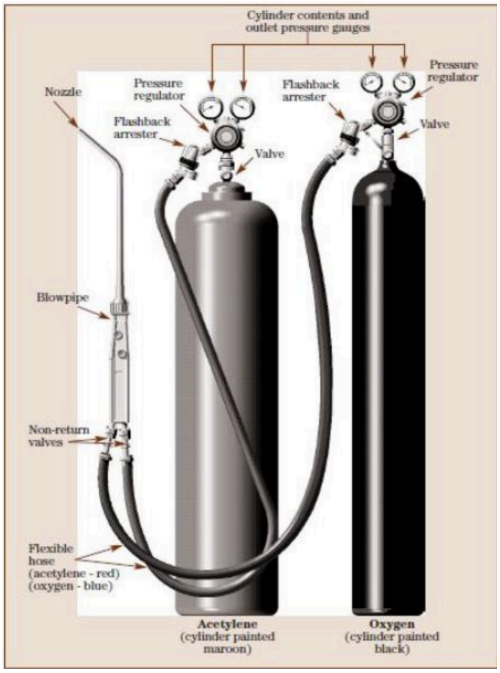
- Flame is formed by mixing oxygen and acetylene in correct proportions in the welding torch.
- The ignited mixture produces an oxy-acetylene flame at the nozzle tip.
- The flame temperature reaches approximately 3200°C, sufficient to melt all commercial metals.
- Allows complete bonding between the metals being joined.

Oxy-Acetylene Equipment:

- It consists of two large steel cylinders:
    - One for oxygen (high pressure)
    - One for dissolved acetylene (high pressure)

- Additional components:
    - Rubber tubes
    - Pressure regulators
    - Blow torch

- Oxygen and acetylene are fed separately to the blow torch, mixed inside, and exit through the nozzle to form the flame.

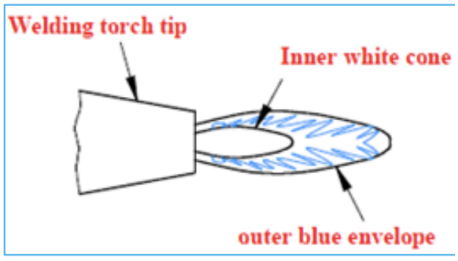


Question:
List and explain the types of oxy-acetylene flames used in gas welding.

Answer:

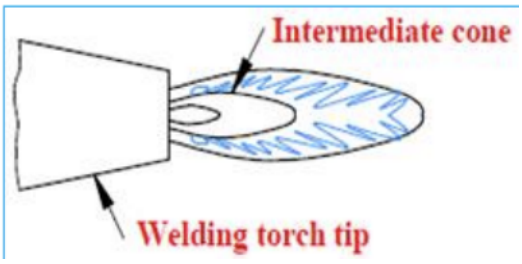
1. Neutral Flame:

- Formed by mixing **equal volumes of oxygen and acetylene** (Oxygen : Acetylene = 1:1).
- Flame structure:
  - Inner small **whitish cone**
  - Surrounded by a sharply defined **blue flame**
- Most commonly used flame in **oxy-acetylene welding**.



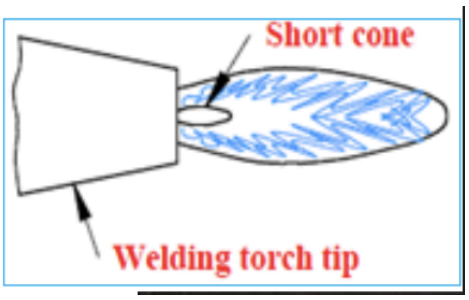
2. Carburizing Flame (Reducing Flame):

- Formed by supplying **excess acetylene** (Oxygen : Acetylene = 0.95:1).
- Flame structure:
  - **Three cones**:
    - Inner white cone
    - Intermediate whitish cone
    - Bluish envelope flame
- Used for welding **alloy steels**, **cast iron**, and **aluminium**
- Protects metal from **oxidizable elements**



3. Oxidizing Flame:

- Formed by supplying **excess oxygen** (Oxygen : Acetylene = 1.5:1).
- Appearance:
  - Similar to neutral flame
  - **Shorter inner white cone**
- Used for welding **nickel** and **non-ferrous materials**
- Commonly used for **oxy-acetylene cutting**, not recommended for welding since it can cause **oxidation of weld metal**



Question:
List and explain common weld defects.

Answer:

1. **Lack of fusion:**
    - Caused by insufficient heat input and excessively fast movement of the welding torch.
    - Results in poor bonding between the weld and base metal.
2. **Porosity:**
    - Occurs when gases become trapped in the weld metal during solidification.
    - Leads to weak spots in the weld.
3. **Inclusions:**
    - Happen when slag is not fully removed before performing another weld pass.
    - Non-metallic materials get embedded in the weld, weakening the joint.
4. **Cracking:**
    - Caused by thermal shrinkage during cooling of the weld.
    - Can severely affect weld integrity.
5. **Undercut:**
    - Occurs due to excess melting of the parent metal along the weld edge.
    - Reduces the strength of the joint by creating a groove.

Question:
Define soldering and list its key features and tools used.

Answer:

- Soldering is a process where two or more metal items are joined by melting and flowing a filler metal (solder) into the joint.
- The filler metal has a lower melting point than the base metals (typically between 150°C to 350°C).
- Unlike welding, soldering does not melt the workpieces.
- Zinc chloride is used as a flux to clean the surfaces and prevent oxidation.
- Heat is applied using a soldering iron, which operates from an electrical source.

<u>Types of Solder:</u>

- Soft solder:
    - Alloy of lead and tin
    - Melting temperature: 150°C to 350°C

    - Used for sheet metal work, plumbing, and electrical junctions
- Hard solder:
    - Alloy of copper, tin, and silver
    - Melting temperature: 600°C to 900°C
    - Used for stronger joints

<u>Tools Used in Soldering:</u>

1. Soldering iron
2. Soldering station
3. Iron tips
4. Brass or conventional sponge
5. Soldering iron stand
6. Solder
7. Helping Hand (device with clips, often includes magnifying glass or light)

Question:
Explain the method of soldering step by step.

Answer:

1. Cleaning of joining surfaces:

- Surfaces are mechanically cleaned to remove dust, oil, scales, etc.
- Ensures proper wetting of surfaces by the molten filler metal.

2. Application of flux:

- Surfaces are coated with flux (commonly rosin or borax).
- Flux chemically cleans the surfaces and promotes bonding with the solder.

3. Tinning of soldering iron:

- The soldering iron is tinned before use.
- Tinning removes any oxide layer on the copper bit and improves heat transfer.

4. Heating:

- The soldering iron is heated, and the molten filler metal is applied to the joint.
- The filler flows into the joint interface and is allowed to cool and solidify, forming the joint.

5. Final clean-up:

- After soldering, the area is cleaned with steel wool or solvent to remove residual flux.
- The soldering iron is cleaned using a damp sponge.

Question:
Explain the brazing process, its key characteristics, and its industrial significance.

Answer:

Brazing Process:

- Brazing is similar to soldering but uses a high-temperature filler metal to join two metals.
- Like welding, brazing forms a mechanical bond between the parts.
- It involves melting, flowing, and solidifying the filler metal to form a leak-tight seal or strong structural bond.
- The brazing filler metal has a liquidus temperature above 450°C but below the solidus temperature of the base metals.
- The base metals are not melted; only the filler metal is melted to create the joint.

Key Characteristics:

- Filler metal melts and flows to form the joint; on solidifying, it joins the parts.
- Filler has a lower melting point than the base metals.
- Dissimilar metals (e.g., aluminium, silver, copper, gold, nickel) can be joined.
- Flux (e.g., borax) is used to: Remove oxides, improve filler flow, promote surface wetting
- Brazed joints can be strong, but not as strong as welded joints.
- Common filler materials include copper alloys, silver alloys, and aluminium alloys.

Industrial Significance and Features:

- Suitable for joining most metallic and ceramic materials.
- Can be performed manually or automatically.
- Ideal for large assemblies and parts made of dissimilar metals.
- Produces a clean, small fillet, unlike the irregular weld bead—useful where appearance matters.
- Offers cost-effective solutions, especially in high-volume production.
- Adaptable to both mass production and single-piece manufacturing.

Question:
Compare welding, soldering, and brazing based on key factors.

Answer:

Factor	Welding	Soldering	Brazing
1. Strength	Strongest joint; can bear high loads. Strength exceeds base metal.	Weakest joint; not load-bearing. Used mainly for electrical work.	Moderate strength; stronger than soldering but weaker than welding.

2. Temperature	Requires very high temperature (~3800°C).	Up to 450°C.	Up to 600°C.
3. Heating of Workpieces	Workpieces are heated to their melting point.	Workpieces are not melted.	Workpieces are heated below their melting point.
4. Mechanical Properties	Base metal properties may change due to intense heat and cooling.	No change in mechanical properties.	Slight change may occur, but generally negligible.
5. Heat Treatment	Often required to remove welding defects.	Not required.	Not required.
6. Preheating Requirement	Not required due to high working temperature.	Preheating recommended for better joint quality.	Preheating desirable to ensure strong joints at lower temperature.

Previous Year Questions:

Explain the various types of computer vision applications in manufacturing.
What are the key components of a computer vision systems architecture, including hardware and software.

Compare computer vision and artificial intelligence in manufacturing.	08

Explain the importance of human vision in computer interaction in manufacturing process.

With necessary diagram, explain the working principle of gas welding process.	

Explain any four defects which are occurred during welding process.	04
List out the types of flames in gas welding and explain with neat diagram.	04

What is arc welding? Explain the fundamental principle behind arc welding.	08
Explain four differences between Welding and Soldering.	08

How do you carry out the Electric arc welding process? Explain with neat process sketch and the safety measures while executing.

In oxy acetylene welding, explain the proportionality of each gas with respect to the type of flames and its application.

With a neat diagram, explain in detail Arc welding process. Name any four applications of arc welding process.	08
Explain with neat diagrams, different types of flames obtained in Oxy-Acetylene flames.	05
Write brief note on welding defects.	03